

# JIECHENG SONG

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## EDUCATION

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### Stony Brook University

PhD in **Statistics**, School of Applied Mathematics and Statistics

August 2022

### Stony Brook University

Master in **Applied Math and Statistics**, School of Applied Mathematics and Statistics

December 2019

### Fudan University

Bachelor in **Information and Computing Science**, School of mathematical sciences

Minor in **Economics**, School of Economics

July 2017

## WORKING EXPERIENCES

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### Software Engineer, Machine Learning

Meta Platforms Inc

Nov 2023 - Present

Video Recommendation

- Contribute development and enhancement of a **Two Tower Sparse Neural Network (TTSN)** reranker **machine learning model** for entity-to-item video recommendation generators for **Facebook Reels Platform**, focusing on leveraging user engagement data and user interest taxonomy.
  - Implemented strategies to train models using specific user interaction data across various video platforms, resulting in significant improvements in model performance.
  - Enhanced the TTSN reranker model by integrating features related to interest taxonomy and user interaction history.
  - Achieved measurable improvements in performance metrics: increased Daily Active Users (DAU) on Facebook Reels by 0.06% and boosted the click-through rate by 0.15%.
  - Successfully increased the prevalence of interest generator by approximately 30%, contributing to enhanced user engagement and content relevance.

### Senior Data Scientist

Wal-mart Associates Inc

March 2023 - Nov 2023

Marketing Decision Science

- **Guide a team of 5 contractors** collaboratively develop and enhance a **tree based predictive machine learning model** to predict customer RFM segment transition, aim at enhancing customer targeting for diverse campaigns.
  - Execute extraction of customer transaction data (over 100M customers and 2B transaction records) from **Google Cloud Service (GCS)**, and formulated customer attributes utilizing **SQL**, **PySpark** and **Pandas** for model training and thorough analysis.
  - Train a predictive machine learning model for customer transitions among RFM segments with **LightGBM**. Employed **undersampling** techniques to address data imbalance, leading to a recall improvement of over 40%. **MLlib** and **H2O** and **scikit-learn** packages was employed in the training process.
  - Construct a weekly schedule data pipeline, incorporating both model training and scoring processes as an **Airflow DAG**. Successfully deploy this pipeline on a **GCP VM** for streamlined operations.
- Develop a **machine learning clustering model** grounded in basket analysis, designed to categorize customers according to their shopping preferences. This serves as a pivotal resource for the marketing team, aiding in precise customer targeting and the formulation of effective marketing strategies.
  - Retrieve customer transaction records and generated customer basket attributes from the transaction records.
  - Execute comprehensive exploration and utilization of various clustering algorithms including **k-means**, **bisecting k-means**, **DBSCAN**, **GMM** models on customer basket data with **scikit-learn** package.
  - Employ the **Elbow method** and **Silhouette Score** techniques to determine the optimal number of clusters, contributing to the informed decision-making process regarding cluster count selection.

### Applied Scientist

Amazon.com Services LLC

September 2022 - March 2023

Customer Engagement Technology

- Work in a team to re-train and re-structure a **classification** model on top of a **transformer based NLP model GPT-2** for customer intent detection with **Tensorflow** and **PyTorch**.
- Contribute to the development and implementation of a **deep learning natural language processing (NLP)** model with hierarchical attention structure for contact summarization, which aims to generate dialogue summaries between customers and bots for customer service agents to save agents' time reading chat history and improve the handling efficiency of agents.

- Retrieve and handle large datasets (Over 2M transcript records) on **AWS RedShift, S3** using **SQL, PySpark** and **Pandas** for model training; cooperate with the data analyst team to prepare gold standard data for evaluation.
- Evaluate the model performance on the gold standard dataset and the new model increases precision by over 30%.
- Built **Docker** containers and images for model inference jobs on **Amazon Sagemaker**.
- Develop a pipeline for dynamically adjusting hyperparameters and thresholds in the touch-button generation function for the customer service chatbot system.
  - Retrieve logging data from **AWS CloudWatch** with AWS object-oriented API for python **Boto3**.
  - Apply **Grid search** strategy to find the best hyperparameters and thresholds for the model in history logging data.
  - Perform **A/B testing** to evaluate the new strategy and the new strategy increases the precision by 12%.

#### Applied Scientist Intern

Amazon.com Services LLC (remote at: 13 Delaware Ct, Coram, NY 11727)

June 2021 - August 2021

Customer Engagement Technology

- Implemented an optimization strategy utilizing **Pareto optimal solution** for touch-button generation function in the **customer service chatbot system**, which is build on top of a transformer based NLP model **GPT-2** for customer intent detection with **Tensorflow** and **PyTorch**.
  - Applied dimension reduction techniques such as **PCA** and **t-SNE** and clustering methods like **k-means** and **DBSCAN** to customer service dialogue data to gain insights and identify patterns.
  - Evaluated and visualized the strategy performance with **Pandas, Numpy, Scikit-Learn** and **Matplotlib**.

## PUBLICATIONS

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Zhuang, Yingying, Song, Jiecheng et al. "Self-supervised pre-training and semi-supervised learning for extractive dialog summarization." Companion Proceedings of the ACM Web Conference 2023. 2023.

Song, Jiecheng, et al. "Data driven pathway analysis and forecast of global warming and sea level rise." Scientific Reports 13.1 (2023): 5536.

Song, Jiecheng, Guanchao Tong, and Wei Zhu. 2023. "A Detecting System for Abrupt Changes in Temporal Incidence Rate of COVID-19 and Other Pandemics" Stats 6, no. 3: 931-941.

Song, Jiecheng, and Merry Ma. "Climate Change: Linear and Nonlinear Causality Analysis." Stats 6.2 (2023): 626-642.

[Google Scholar link](#)

## SKILLS

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|---------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Programming Languages:</b>   | Python, R, SQL, MATLAB, C/C++, SAS                                                                               |
| <b>Machine Learning Tools:</b>  | Pandas, Numpy, Scikit-Learn, Pyspark, PyTorch, Tensorflow, Transformers and etc.                                 |
| <b>Machine Learning Skills:</b> | Regression, Clustering, Classification, Neural Network, Natural Language Processing (NLP), Recommendation System |
| <b>Developer Tools:</b>         | Git, Airflow, VS Code, Docker, AWS, GCP                                                                          |